

WorldCrafter: Consistent Video World Model with Implicit 3D-aware Memory

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<https://drexubery.github.io/WorldCrafter>

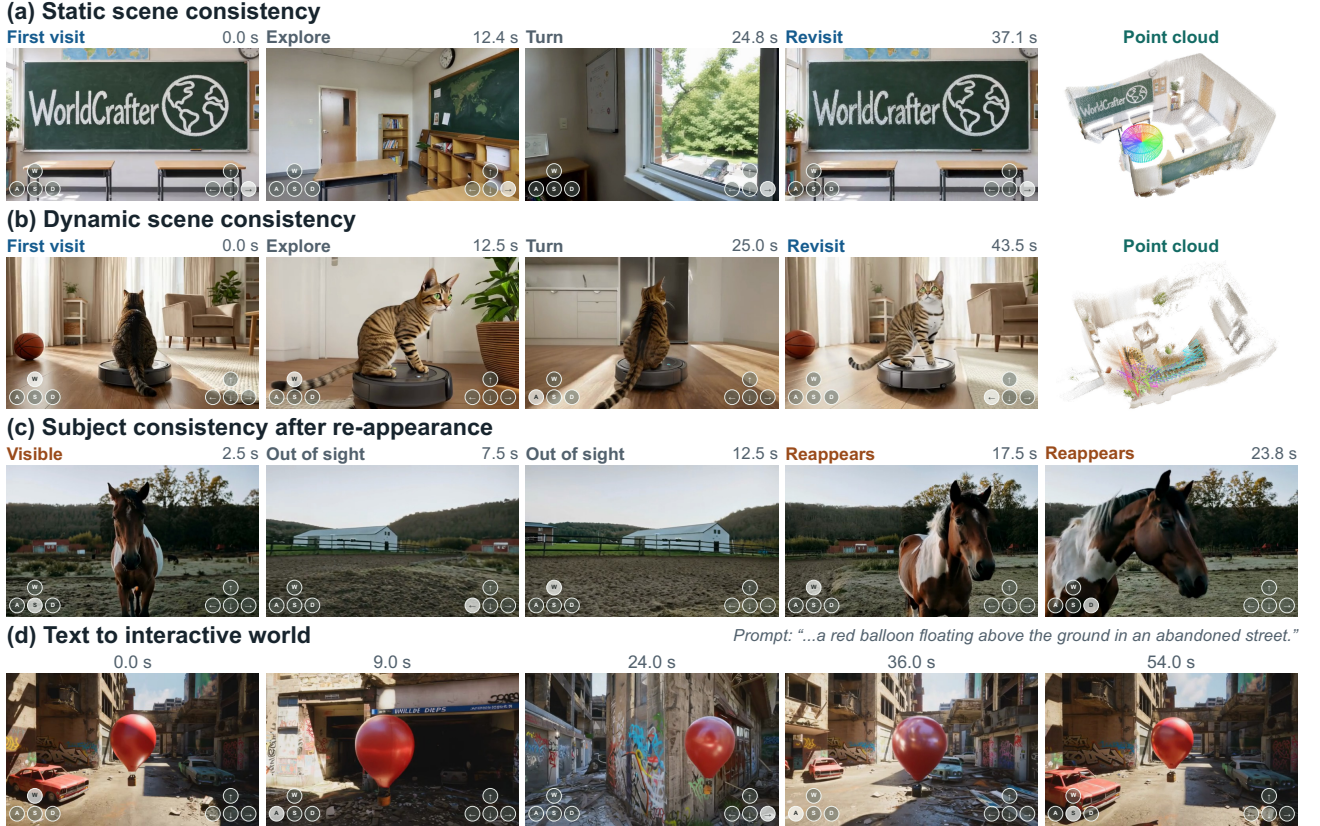


Figure 1. WorldCrafter enables consistent, camera-controlled exploration. It preserves scene appearance and structure across revisits in static (a) and dynamic (b) scenes, as illustrated by reconstructed point clouds. Subjects remain consistent after leaving and re-entering view (c). Users can also generate and explore scenes from text descriptions alone (d).

Abstract

Video world models enable interactive exploration of dynamic environments, yet struggle to respect prior observations over long horizons and across viewpoints. We present WorldCrafter, a video world model that learns a camera-queryable implicit 3D-aware memory for this purpose. The key insight is to let the requested viewpoint shape how multi-view evidence is compressed into the video generator’s limited token budget. Trained jointly with the video genera-

tor, a memory encoder and pose-conditioned readout module integrate historical observations into a fixed set of target view-specific tokens before denoising, without explicit depth-based correspondences. By combining this memory with recent temporal context and few-step distillation, WorldCrafter enables streaming scene exploration from a single input image or text prompt. Experiments across static and dynamic scenes show substantial gains in long-horizon consistency and camera-control accuracy while preserving visual quality during minute-scale exploration.

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1. Introduction

Video world models enable interactive exploration of dynamic environments by generating new observations as users move the camera [15, 19, 36, 47, 50, 51, 65, 73, 104]. Maintaining a coherent world requires memory beyond the recent context so that previously observed content remains consistent when revisited, even from a different viewpoint.

A straightforward way to provide memory for video world models is to include previously generated frames in attention. Full-history attention incurs substantial computation costs [9, 23, 84], while selective history retrieval trades view coverage for efficiency [9, 12, 38, 60, 62, 73, 75, 79, 84, 89]. Explicit spatial memories provide a shared 3D reference but depend on accurate geometry and struggle with dynamic scenes [76, 88, 94, 101]. Implicit memories compress history into learned representations [59, 95], with recent methods incorporating geometry features for 3D awareness [26, 68, 74]. However, these geometry-oriented representations prioritize geometric prediction over the appearance fidelity needed to reproduce previously observed scenes [63].

Recent advances in 3D representation learning [31–33, 63] have demonstrated remarkable capabilities in learning compact scene representations, making them a natural foundation for the memory space of video world models. Motivated by this, we present **WorldCrafter**, a video world model with implicit 3D-aware memory for consistent interactive generation. At its core, a memory encoder initialized from pretrained 3D representation encoders maps historical latent frames into a compact memory space, inheriting their learned 3D inductive bias. We further optimize this memory space by jointly training the memory encoder, video diffusion transformer (DiT), and a memory readout module, enabling the memory to co-adapt with the DiT token space.

To extract generation-relevant information from this memory, we compare pose-free and pose-guided readout under a fixed token budget. Pose-guided readout focuses on information relevant to the requested viewpoints and yields better revisit consistency and camera control in our experiments. The resulting tokens condition the DiT directly through self-attention, without reconstructing target-view images. During interaction, we select complementary historical views according to their joint camera coverage and combine the queried memory with recent temporal context. The memory supplies historical scene information, while recent context supports the continuation of visible motion. Generated chunks are added to the history archive, while the encoder input size and DiT memory-token budget remain fixed. Together with camera-conditioned autoregressive generation and few-step distillation, our model achieves real-time streaming inference while preserving precise camera control and minute-scale consistency under complex user-specified camera trajectories.

Our contributions are as follows:

- We introduce an implicit 3D-aware memory mechanism for video world models. It learns to encode history latent frames into a compact memory representation that preserves spatial-temporal context, enabling efficient memory writing and readout within a fixed token budget.
- We integrate this memory mechanism into camera-controllable autoregressive video generation, achieving leading revisit consistency (47.6% improvement relative to the strongest baseline) and camera-control accuracy, with controlled ablations supporting our design.
- We build a real-time interactive system through few-step distillation, achieving streaming inference while maintaining visual quality throughout minute-scale exploration.

2. Related Work

2.1. Interactive Video World Models

Video world models generate future observations in response to user actions, turning video generation into interactive simulation [2, 8, 27, 67]. Recent methods increasingly combine streaming generation with camera control to support real-time interaction and long-horizon exploration [1, 14, 16, 18, 19, 25, 30, 47, 48, 50, 55, 60, 62, 73, 88, 100, 104].

Streaming generation extends video diffusion through rolling denoising or temporally varying noise levels [10, 11, 34, 57, 58], while autoregressive distillation and rollout-aware training improve sampling efficiency and mitigate error accumulation [28, 45, 87, 103, 105]. Flexible history conditioning supports longer rollouts [22, 61], complemented by efficient streaming designs and parallel implementations [13, 82, 102]. However, extending temporal rollouts alone does not ensure faithful recall of previously observed scenes.

Camera control is commonly implemented through discrete action inputs [15, 30, 36, 46, 73, 90], continuous camera parameters [4–6, 20, 21, 39, 71, 80, 97], or point-cloud renders along the target trajectory [56, 92, 93]. Although these signals specify viewpoint changes, they do not provide a persistent state that preserves scene content beyond the context window. Thus, combining streaming generation with camera control alone remains insufficient for consistent long-horizon exploration.

2.2. Memory Mechanisms in Video World Models

Persistent video world models require memory beyond the recent video context. We categorize existing approaches according to their stored representations: context memory, spatial memory, and implicit memory.

Context memory retains historical frames, latent tokens, or cached attention features for attention-based reuse [9, 14,

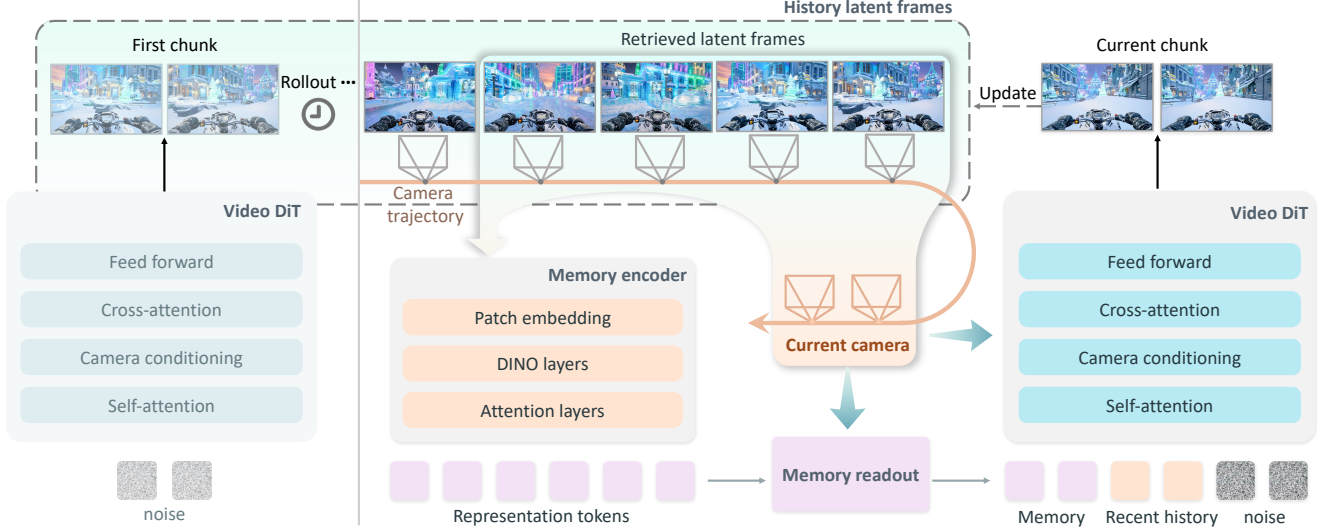


Figure 2. **Overview of the WorldCrafter pipeline.** The video DiT generates the first chunk conditioned on camera poses, before any history is available. At subsequent rollout steps, max-coverage history retrieval selects latent frames based on the current camera poses, and the memory encoder maps them into a compact 3D-aware representation. A pose-conditioned memory readout module extracts fixed-size memory tokens that, together with recent history, guide generation of the current chunk. Generated frames are added to the history for subsequent rollouts.

23, 49, 62, 73, 78, 79, 89, 98]. Frame-retrieval methods select a small subset using camera overlap or reconstructed surfaces [17, 24, 38, 89], while learned querying can aggregate information across the available history. MemLearner [91] uses query tokens that attend to both historical context and noisy predictions in shallow DiT layers; deeper layers consume the queries without the original context tokens. CaR [53] compresses historical latents and retrieves them through relative-camera attention inside the denoising network. These methods learn to access historical context within the generator. WorldCrafter instead aggregates history into a 3D-aware memory representation using a pre-trained multi-view encoder, then reads out fixed-size memory tokens before denoising.

Spatial memory instead transforms history frames into views specified by target camera poses [37, 56, 72, 76, 81, 88, 93, 94, 101]. Representative methods perform this transformation via novel view synthesis, encode the synthesized target-view frames into the VAE latent space, and concatenate the resulting latents channel-wise with the input noise to condition video diffusion models [37, 56, 72, 76, 93]. The resulting spatial correspondence across viewpoints facilitates revisiting previously observed regions. However, strong alignment to the target view can overconstrain scene dynamics, limiting their ability to model dynamic objects and environments.

Implicit memory encodes history into learned representations [12, 40, 59, 75, 95]. Existing methods update memory recurrently alongside local context [54, 59, 77, 95] or compress observations using learned encoders [12, 75].

Geometry-aware approaches draw on pretrained geometry estimators such as VGGT [68]: CineScene [26] uses their features as generation conditions, while GIM-World [74] distills them into memory through geometric supervision. However, geometry-estimation pretraining prioritizes geometric prediction over the appearance fidelity needed for consistent visual recall [63]. WorldCrafter instead adapts multi-view scene representations learned through novel-view reconstruction, which requires preserving both geometry and appearance. Initialized from LagerNVS [63], our memory encoder and readout inherit a learned 3D inductive bias and are jointly optimized with the video generator.

3. Method

3.1. Preliminary

Latent video diffusion. Let $\mathbf{x} \in \mathbb{R}^{3 \times F \times H \times W}$ denote a clean video of F frames. A video variational autoencoder (VAE) [35] maps it to a spatiotemporal latent $\mathbf{z} = \mathcal{E}_{\text{VAE}}(\mathbf{x}) \in \mathbb{R}^{c \times f \times h \times w}$. Its decoder reconstructs the video as $\hat{\mathbf{x}} = \mathcal{D}_{\text{VAE}}(\mathbf{z})$. Operating in this latent space substantially reduces the token sequence processed by the Diffusion Transformer (DiT) [52]-based denoiser, which patchifies $\mathbf{z}$ into video tokens and applies 3D self-attention. In a conventional bidirectional video diffusion model such as Wan 2.1 [64], every video token can attend to all other tokens in the clip during each denoising step.

The denoiser is trained with conditional flow matching [44]. For diffusion time $t \sim \mathcal{U}(0, 1)$ and Gaussian noise

$\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, the noisy latent and its target velocity are

$$\mathbf{z}_t = (1 - t)\mathbf{z} + t\epsilon, \quad \mathbf{v}_t^* = \epsilon - \mathbf{z}. \quad (1)$$

Given a text condition $\mathbf{y}$, the DiT $\mathbf{v}_\theta$ predicts this velocity by minimizing

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{\mathbf{z}, t, \epsilon} \left[\|\mathbf{v}_\theta(\mathbf{z}_t, t \mid \mathbf{y}) - \mathbf{v}_t^*\|_2^2 \right]. \quad (2)$$

Chunk-wise autoregressive video generation. To extend generation beyond a fixed clip, autoregressive methods divide a long video into fixed-length chunks and generate them sequentially. At a generic rollout step, we omit the rollout index from all quantities for clarity. Let $\mathbf{z}$ denote the clean latent of the current chunk, $\mathbf{z}_t$ its state at diffusion time t , $\mathbf{z}^h$ the accumulated clean history frames, and $\mathbf{z}^r$ the fixed-length recent history frames retained from $\mathbf{z}^h$. Given the text condition $\mathbf{y}$, a standard chunk-wise autoregressive model evolves the current latent through the conditional flow

$$\frac{d\mathbf{z}_t}{dt} = \mathbf{v}_\theta(\mathbf{z}_t, t \mid \mathbf{z}^r, \mathbf{y}). \quad (3)$$

At each denoising step, the DiT processes the concatenated latent sequence $[\mathbf{z}^r; \mathbf{z}_t]$, where $[\cdot; \cdot]$ denotes concatenation along the token sequence. After denoising, $\mathbf{z}$ is appended to $\mathbf{z}^h$, and the sliding window of $\mathbf{z}^r$ is updated with $\mathbf{z}$.

3.2. Model Architecture

To enable long-horizon memory and user interaction, we extend the standard autoregressive flow in Eq. (3) by conditioning each chunk jointly on a memory $\mathbf{M}$ derived from the accumulated history $\mathbf{z}^h$ and a target camera trajectory $\mathbf{C}$:

$$\frac{d\mathbf{z}_t}{dt} = \mathbf{v}_\theta(\mathbf{z}_t, t \mid \mathbf{M}, \mathbf{z}^r, \mathbf{C}, \mathbf{y}). \quad (4)$$

The overview of our pipeline is shown in Figure 2. The video DiT generates the first chunk conditioned on camera poses alone, as no history is yet available. As history accumulates, we learn a memory encoder to map history latent frames into a compact 3D-aware representation, which is read out as a fixed-size memory $\mathbf{M}$ to condition subsequent generation.

Memory conditioning. At each denoising step, the DiT processes $[\mathbf{M}; \mathbf{z}^r; \mathbf{z}_t]$ as a single latent sequence. This adds a dedicated memory stream to the recent history while preserving chunk-level causality. Unlike prior context-based memory methods [62, 73, 79, 89] that instantiate $\mathbf{M}$ as a fixed-length context retrieved from the accumulated history $\mathbf{z}^h$, we model $\mathbf{M}$ by mapping $\mathbf{z}^h$ into a compact 3D-aware implicit memory representation.

Camera conditioning. The target trajectory $\mathbf{C}$ specifies a camera-to-world pose and camera intrinsics for each frame. Following PROPE [39], we encode relative camera geometry as a positional transformation within self-attention.

We implement this conditioning using the parallel camera-attention branch of UCPE [97], which adopts independent query, key, and value projections and adds its output to the original self-attention through a zero-initialized projection. The camera branch is applied only to the noisy part $\mathbf{z}_t$ of the concatenated sequence; the memory $\mathbf{M}$ and recent context $\mathbf{z}^r$ are processed without camera injection.

3.3. Memory Encoder

Memory writing. As shown in Figure 2, after generating the first chunk, the memory encoder Φ writes the accumulated history latents $\mathbf{z}^h$ and their corresponding camera parameters $\mathbf{C}^h$ into an implicit 3D-aware representation $\mathbf{R}$:

$$\mathbf{R} = \Phi(\mathbf{z}^h, \mathbf{C}^h) \in \mathbb{R}^{|\mathbf{z}^h|L \times d}, \quad (5)$$

where $|\mathbf{z}^h|$ denotes the number of history latent frames. The encoder produces L tokens of dimension d per history latent frame.

We initialize the encoder architecture and weights from the LagerNVS encoder [63], discarding its shallow image-processing layers and adding a new patch embedding layer to map each latent frame directly into its representation space. The history camera poses are expressed relative to the latest latent frame in $\mathbf{z}^h$ and injected as camera tokens during encoding. The resulting representation tokens $\mathbf{R}$ aggregate geometry and appearance information across the input history without materializing an explicit 3D reconstruction.

Since the length of $\mathbf{R}$ grows linearly with the number of input history frames $|\mathbf{z}^h|$, to bound the cost of the encoder, we restrict its input to k history latent frames. At inference, we retain the latest latent frame in $\mathbf{z}^h$ and greedily select $k-1$ complementary frames whose joint field of view (FoV) maximizes coverage of the target region along the upcoming camera trajectory. We denote the selected history latents and their camera parameters by $\mathbf{z}^s$ and $\mathbf{C}^s$, respectively. Prior context-based memory methods [62, 73, 79, 89] retrieve only a few history frames by ranking their pairwise FoV similarity to target poses [73, 89], resulting in limited coverage and sensitivity to individual selections. In contrast, our memory encoder accommodates more history frames under a comparable budget, while max-coverage history retrieval yields broader coverage and greater robustness.

Memory readout. The written representation $\mathbf{R}$ should then be read out as a fixed-size memory $\mathbf{M}$ that conditions the DiT. We compare two readout mechanisms under a fixed DiT token budget.

The first is pose-free readout, in which a learned readout module maps the complete representation into a fixed set of memory tokens compatible with the DiT input:

$$\mathbf{M} = \text{Readout}(\Phi(\mathbf{z}^s, \mathbf{C}^s)). \quad (6)$$

This readout is independent of the upcoming camera trajectory, leaving the DiT attention to identify information relevant to the current generation.

The second is pose-guided readout, which queries the representation using a fixed-size set of query poses $\mathbf{C}^q \subset \mathbf{C}$ sampled from the upcoming target camera trajectory:

$$\mathbf{M} = \text{Readout}(\Phi(\mathbf{z}^s, \mathbf{C}^s), \mathbf{C}^q). \quad (7)$$

In pose-guided readout, we initialize the readout module with the shallow decoder layers of LagerNVS [63] and add projection layers to map the output tokens into the DiT token space.

Empirically, we find that pose-guided readout outperforms pose-free readout and yields more accurate camera control. We attribute this gain to a more effective allocation of the fixed memory budget to target-relevant information. We therefore adopt pose-guided readout, with ablations presented in Sec. 4.5.

3.4. Base Model Training

Dataset curation. Our training data combine the OpenSora-Plan (OSP) dataset [41], DL3DV [43], and synthetic videos from MIND [83], covering diverse indoor and outdoor scenes and object motions. We use Depth Anything 3 [42] to obtain metric-scale camera pose annotations across all data sources and Qwen2.5-VL [7] to generate video captions. Using these captions and the estimated camera trajectories, we further curate a subset of the OSP dataset in which the camera follows moving subjects, helping the model learn coordinated camera and subject motion.

Training details. We initialize the video DiT from Helios-base [96] and train our base model in 4 stages. The original Helios-base inference window contains a 9-frame noise chunk and a FramePack-style clean history [98] comprising a compressed 16-frame segment, a 2-frame segment, the latest latent frame, and an attention-sink frame. We remove its compressed 16-frame segment and prepend memory tokens equivalent in number to the tokens of 4 uncompressed history frames.

In the first stage, we fine-tune Helios-base to adapt to our modified inference window. For each sampled video chunk, we randomly select 4 history frames from the preceding 4 chunks to populate the memory slots. We train on 760,000 videos from the OSP dataset for 5,000 iterations using 32 GPUs with a global batch size of 32.

In the second stage, we introduce camera control by training the UCPE-based camera conditioning branch while keeping the video DiT backbone frozen. We use 40,000 videos from the filtered OSP subset and 6,000 videos from DL3DV, training on 32 GPUs with a global batch size of 128.

In the third stage, we adapt our memory encoder to process VAE latents. We initialize it from the LagerNVS en-

coder and replace its shallow DINO layers with a latent patch embedding layer, as described in Sec. 3.3. The encoder takes a fixed number of 9 latent frames as input. We warm up the memory encoder on DL3DV and the filtered OSP subset for 5,000 iterations using 16 GPUs with a global batch size of 16.

In the final stage, we introduce a memory readout module that outputs a fixed number of memory tokens matching the token count of 4 full frames. We jointly train this module with the memory encoder, video DiT, and camera conditioning branch to co-adapt the learned memory representation and the DiT token space. We first train on DL3DV and the filtered OSP subset for 8,000 iterations using 32 GPUs with a global batch size of 32, then incorporate synthetic videos from MIND for 1,000 additional iterations to improve dynamic subject modeling.

3.5. Distillation for Real-time Interaction

Pyramid distillation. Following Helios [96], we adopt a coarse-to-fine pyramid denoising scheme and apply distribution matching distillation [85, 86] to reduce the number of sampling steps. We use 3 spatial resolutions with 2 denoising steps per resolution. To support camera conditioning across the pyramid, we rescale the spatial coordinates while keeping the camera poses and field of view unchanged across pyramid levels during UCPE camera embedding.

Hybrid distilled model. We observe a trade-off between visual fidelity and subject-following ability when distilling with synthetic data. Incorporating synthetic data [83] improves the model’s subject-following ability, but can also introduce smeared textures. To preserve both subject-following ability and natural visual details, we distill a low-noise model and a high-noise model with different training data compositions. The low-noise model is distilled from the base model before synthetic data adaptation, using the filtered OSP subset and DL3DV dataset to preserve natural appearance. The high-noise model is distilled from the base model after synthetic data adaptation, using a mixture of OSP, DL3DV, and MIND to retain subject-following ability. During inference, the low-noise model performs the last denoising step, while the high-noise model performs all the preceding steps. After distillation, WorldCrafter-fast can achieve a generation speed of 16 fps on a 4-GPU machine.

4. Experiments

4.1. Experimental Setup

Benchmark. We curate a benchmark to evaluate memory ability, camera-control accuracy, and visual quality in long-horizon video world models. The benchmark contains 145 images from HappyOyster [19], Project Genie [50], web

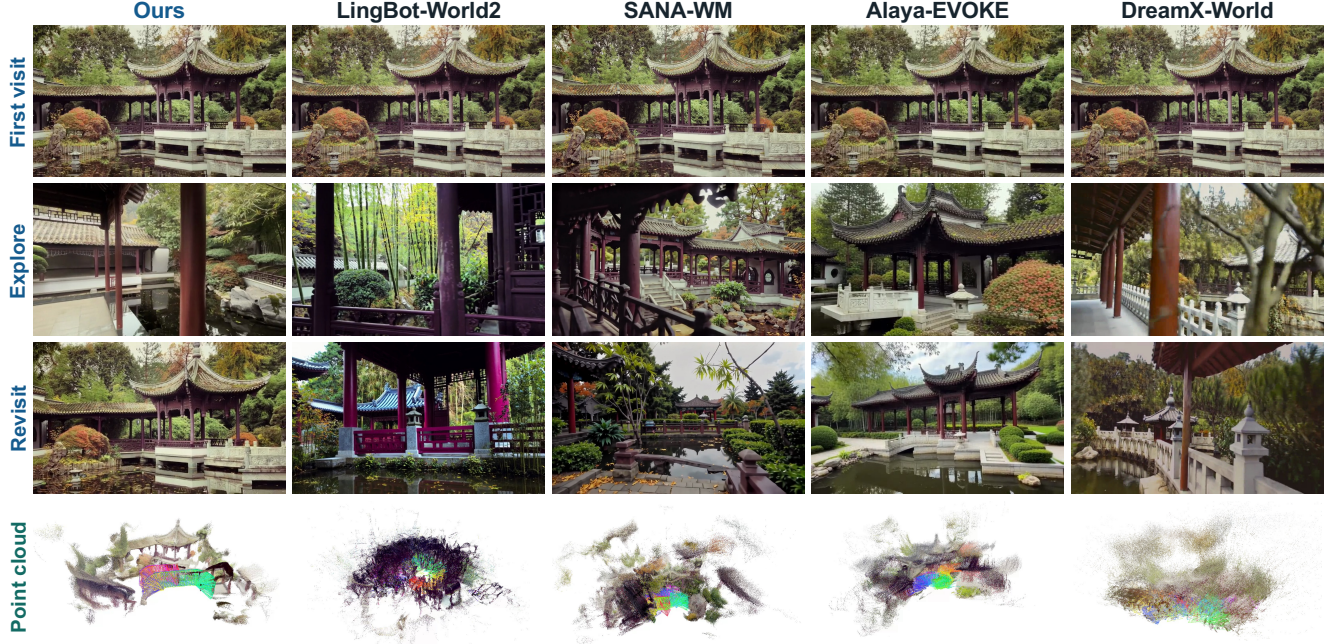


Figure 3. **Qualitative long-horizon revisit comparison in a static scene.** Rows show the first observation, an intermediate view during exploration, the matched revisit, and the point cloud reconstruction of the generated video, respectively.

Table 1. **Long-horizon revisit consistency.** Results are averaged over 725 generated videos using matched first-visit and revisit frames. Progressively lighter shades of blue mark the best, second-best, and third-best results, respectively.

Method	MEt3R↓	LPIPS↓	PSNR↑	SSIM↑
DreamX-World [16]	0.548	0.627	12.898	0.243
Alaya-EVOKE [88]	0.414	0.565	12.332	0.290
HY-WorldPlay [62]	0.394	0.515	12.983	0.252
Lyra 2.0 [60]	0.334	0.487	14.050	0.390
Echo-WM [100]	0.449	0.582	12.592	0.239
LingBot-World 2 [18]	0.492	0.633	10.449	0.219
Matrix-Game 3.5 [55]	0.405	0.549	12.976	0.224
SANA-WM [104]	0.397	0.553	13.142	0.246
WorldCrafter	0.166	0.255	18.016	0.517
WorldCrafter-fast	0.129	0.186	20.868	0.616

sources, and images generated by GPT-Image2, covering 83 dynamic object-centric scenes and 62 static scenes. Each image and its text description are paired with 5 metric camera trajectories, yielding 725 videos per method. The trajectories span 528–1,648 frames and include closed-loop revisits to assess whether previously observed content is preserved over long horizons.

Comparison methods. We evaluate two variants of our method: WorldCrafter and its distilled counterpart, WorldCrafter-fast. We compare our models with 8 re-

Table 2. **Camera-control accuracy.** Results are averaged over 725 generated videos using Sim(3)-aligned estimated and target trajectories. Progressively lighter shades of blue mark the best, second-best, and third-best results, respectively.

Method	RotErr↓	TransErr↓	CamMC↓
DreamX-World [16]	54.116	2.759	3.138
Alaya-EVOKE [88]	26.042	2.042	2.199
HY-WorldPlay [62]	34.051	2.146	2.359
Lyra 2.0 [60]	16.145	1.538	1.624
Echo-WM [100]	21.455	2.072	2.189
LingBot-World 2 [18]	30.615	2.004	2.192
Matrix-Game 3.5 [55]	19.881	1.920	2.028
SANA-WM [104]	23.531	1.740	1.887
WorldCrafter	13.536	1.475	1.546
WorldCrafter-fast	18.251	1.638	1.737

cent camera-controllable video world models: DreamX-World [16], Alaya-EVOKE [88], HY-WorldPlay [62], Lyra 2.0 [60], Echo-WM [100], LingBot-World 2 [18], Matrix-Game 3.5 [55], and SANA-WM [104].

These baselines span different memory representations and camera-conditioning mechanisms. HY-WorldPlay and DreamX-World retrieve history context based on camera similarity and use PROPE [39] for camera control. SANA-WM combines Gated DeltaNet memory with UCPE [97] and Plücker-ray conditioning, whereas Echo-WM employs

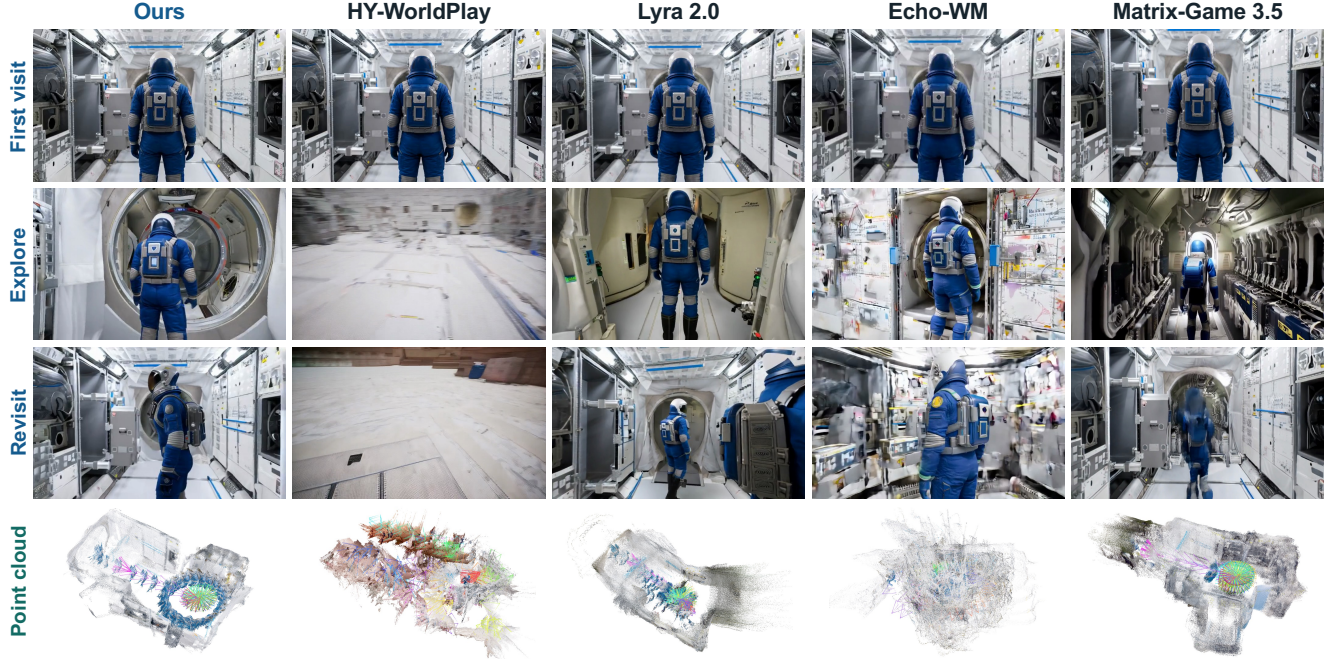


Figure 4. **Qualitative long-horizon revisit comparison in a dynamic scene.** Rows show the first observation, an intermediate view during exploration, the matched revisit, and the point cloud reconstruction of the generated video, respectively.

Table 3. **Visual quality on VBench.** Scores are computed in custom-input mode over 725 generated videos. SC, BC, TF, MS, AQ, IQ, DD, and OC denote subject consistency, background consistency, temporal flickering, motion smoothness, aesthetic quality, imaging quality, dynamic degree, and overall consistency, respectively. The Overall column reports the aggregate VBench score. Progressively lighter shades of blue mark the best, second-best, and third-best results, respectively.

Method	SC	BC	TF	MS	AQ	IQ	DD	OC	Overall
DreamX-World [16]	75.994	87.601	95.327	98.100	57.593	60.412	99.806	24.583	77.652
Alaya-EVOKE [88]	80.522	88.994	95.872	98.481	61.579	71.249	99.029	25.015	81.406
HY-WorldPlay [62]	75.407	86.629	95.553	97.626	53.454	61.700	97.087	21.182	76.539
Lyra 2.0 [60]	77.277	87.935	95.167	98.652	57.832	65.149	99.806	21.484	78.943
Echo-WM [100]	81.671	90.206	95.771	96.751	61.818	66.564	98.835	26.082	80.211
LingBot-World 2 [18]	75.584	87.725	94.206	96.657	60.815	68.776	100.000	25.605	78.174
Matrix-Game 3.5 [55]	70.370	83.626	94.395	98.087	58.615	69.374	100.000	18.813	76.967
SANA-WM [104]	79.413	89.625	95.768	98.608	63.467	66.080	98.835	25.234	80.841
WorldCrafter	82.695	90.589	96.217	98.787	61.432	69.058	96.893	26.745	81.910
WorldCrafter-fast	81.365	89.559	96.444	98.736	58.650	63.982	96.505	24.306	80.285

a UCPE-based camera branch and sliding-window memory. Alaya-EVOKE and Lyra 2.0 construct spatial memory using depth estimated by Depth Anything 3 [42] and condition generation through geometric warping. Matrix-Game 3.5 combines geometric patch memory with Warped PRoPE, using VGGT- Ω [69] and Depth Anything 3 for metric-scale geometry annotation.

We evaluate the full-step base model with a refiner for SANA-WM, the full-step base models for Lyra 2.0 and HY-WorldPlay, and the distilled models for the remaining base-lines. All methods receive identical initial images, text de-

scriptions, and target trajectories. Before evaluation, we re-size the generated videos to 640×384 to ensure a common evaluation resolution.

4.2. Memory Evaluation

Following the protocol in [65], we evaluate memory ability by comparing frames generated upon revisiting a location with the corresponding frames from the initial visit. We report MET3R [3], LPIPS [99], PSNR, and SSIM [70] to assess consistency between the paired observations.

As shown in Table 1, our models achieve the top two

Table 4. **Ablation of the memory design.** Each variant changes one design choice while keeping the training and inference settings and the DiT memory-token budget fixed.

Method	Memory evaluation				Camera evaluation		
	MEt3R↓	LPIPS↓	PSNR↑	SSIM↑	RotErr↓	TransErr↓	CamMC↓
Context memory	0.382	0.497	13.907	0.315	26.522	2.083	2.150
Frozen memory encoder	0.227	0.305	16.873	0.472	15.428	1.793	1.886
Pose-free memory readout	0.251	0.333	16.486	0.467	18.307	1.701	1.828
Similarity-based history retrieval	0.213	0.296	17.125	0.485	14.657	1.579	1.664
WorldCrafter	0.166	0.255	18.016	0.517	13.536	1.475	1.546

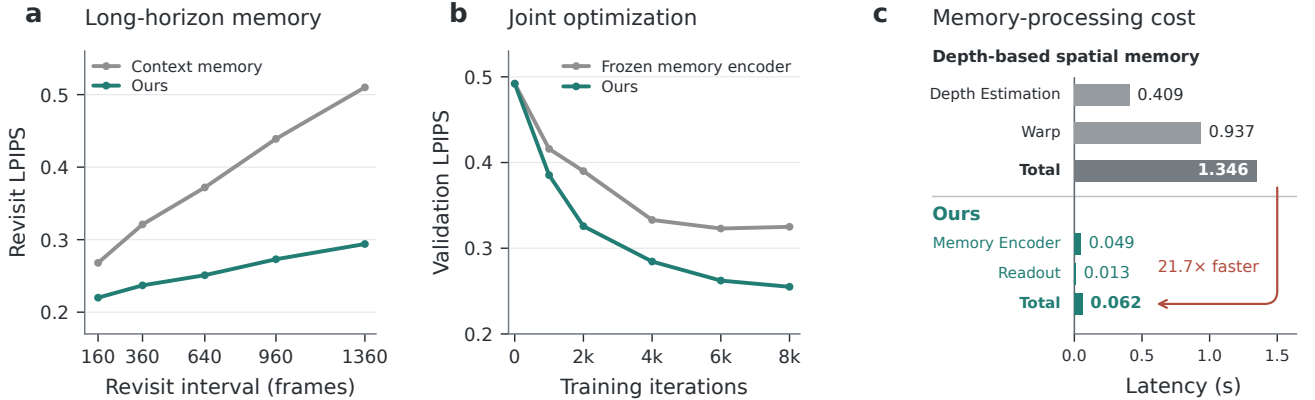


Figure 5. **Ablations of implicit 3D-aware memory, joint optimization, and memory-processing efficiency.** (a) Long-horizon revisit consistency of context memory and WorldCrafter’s implicit 3D-aware memory, measured by LPIPS across revisit intervals. (b) Validation LPIPS for joint optimization and frozen memory-encoder training at matched iterations. (c) Per-chunk memory-processing latency of depth-based spatial memory and WorldCrafter.

results on all 4 metrics, with WorldCrafter-fast performing best. Compared with Lyra 2.0, WorldCrafter reduces LPIPS from 0.487 to 0.255 and increases PSNR from 14.050 to 18.016 dB. The agreement across these metrics indicates that revisited views more faithfully recover the appearance and structure of earlier observations, supporting the effectiveness of the learned memory representation in long-horizon closed-loop exploration. Figures 3 and 4 compare first-visit and revisit frames on static and dynamic scenes, together with point clouds reconstructed from the generated videos using VGGT- Ω [69]. WorldCrafter’s revisit frames closely match the corresponding first-visit observations, while its generated videos yield coherent point clouds with well-aligned camera poses. These qualitative results further indicate that WorldCrafter preserves scene structure and appearance during long-horizon video generation while maintaining visual quality.

4.3. Camera Control Evaluation

We sample the generated videos at a stride of 4 frames and recover camera trajectories using VGGT- Ω [69]. Follow-

ing SANA-WM [104], each trajectory is normalized relative to its first pose and aligned using Umeyama Sim(3) alignment [66]. Evaluation metrics include rotation error (RotErr), camera-center translation error (TransErr), and pose-matrix discrepancy (CamMC), with lower values indicating more accurate camera control. As shown in Table 2, WorldCrafter achieves the lowest error on all 3 metrics, while WorldCrafter-fast ranks third on each metric.

4.4. Visual Quality Evaluation

Following SANA-WM [104], we evaluate video generation quality using VBench [29] in custom-input mode. We report Subject Consistency (SC), Background Consistency (BC), Temporal Flickering (TF), Motion Smoothness (MS), Aesthetic Quality (AQ), Imaging Quality (IQ), Dynamic Degree (DD), and Overall Consistency (OC). As shown in Table 3, our models achieve the best results in 5 of the 8 dimensions. WorldCrafter obtains the highest overall score of 81.910, while WorldCrafter-fast achieves the best Temporal Flickering score. These gains primarily reflect stronger scene consistency and temporal coherence across the gener-

ated videos, complementing the results of the memory evaluation.

4.5. Ablation Study

All ablations are conducted on WorldCrafter before distillation, following the memory and camera-control evaluation protocols in Secs. 4.2 and 4.3. Table 4 compares the full model with 4 variants under the same training and inference settings.

Implicit 3D-aware memory versus context memory. The *context memory* variant replaces the memory tokens produced by the memory encoder and memory readout module with 4 retrieved history latent frames. It is initialized from the second-stage model in Sec. 3.4 and further trained for the same number of iterations as the full model. Table 4 shows that this replacement degrades both revisit consistency and camera control. To examine how memory retention varies over time, Figure 5(a) plots mean revisit LPIPS against the frame interval between the initial observation and its revisit. WorldCrafter exhibits a slower increase in revisit error, with a widening advantage over context memory at longer intervals.

Joint optimization of the memory encoder and video generator. The *frozen memory encoder* variant fixes the encoder during the final training stage, while keeping the memory readout module, video DiT, and camera conditioning branch trainable. This variant yields weaker memory and camera-control performance in Table 4. Figure 5(b) tracks validation performance at matched training iterations: joint optimization reaches lower revisit error earlier and maintains this advantage. This supports co-adapting the memory representation with the video generator instead of learning to consume a fixed representation space.

Memory readout. The *pose-free memory readout* variant maps the encoded representation to memory tokens without target-pose queries, retaining the same history inputs and output token budget. Table 4 shows that pose-guided readout improves both memory and camera control ability.

History retrieval. To isolate the effect of history retrieval, the *similarity-based history retrieval* variant replaces max-coverage retrieval with pairwise FoV similarity ranking. Both variants retain the latest latent frame and select 8 additional history frames as the memory encoder input, with the same DiT memory-token budget. Table 4 shows improved revisit consistency and camera control with max-coverage retrieval. This comparison favors selecting complementary views that jointly cover the target region over independently ranking views by similarity, without increasing the number of history inputs.

Memory efficiency. Depth-based spatial memory methods, including Lyra 2.0 [60], Matrix-Game 3.5 [55], and Alaya-EVOKE [88], rely on estimated geometry to reuse history observations. We compare the additional cost of depth es-

timization and warping with that of WorldCrafter’s memory encoder and readout. Figure 5(c) reports component latencies at 640×384 resolution, with a generation chunk of 9 latent frames and 4 history frames for spatial warping. Shared VAE decoding for RGB output and video denoising are excluded. For spatial memory, the depth estimation and alignment process takes 0.409 s using Depth Anything 3 [42], while the batched warping takes 0.937 s, totaling 1.346 s per chunk. By operating directly on history latents, WorldCrafter requires 0.049 s for memory encoding and 0.013 s for readout, totaling 0.062 s. This reduces the summed memory-processing cost by a factor of $21.7\times$, without requiring explicit depth estimation or warping.

5. Discussion

We present WorldCrafter, a camera-controllable autoregressive video world model with implicit 3D-aware memory. A learned memory encoder aggregates history latent frames into a compact representation, which is read out as tokens compatible with the video DiT. Experiments demonstrate improved revisit consistency and camera-control accuracy over the evaluated baselines.

Several limitations remain. Consistency can still break down along particularly complex or extended trajectories. In addition, re-encoding history at every chunk incurs extra latency. A promising direction is an autoregressive streaming memory encoder that incrementally incorporates each newly generated chunk into the memory state, reducing repeated computation over history.

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